

R for SAS Users

Melinda Higgins

<https://statisticalhorizons.com/seminars/r-sas-users/>

2-day Workshop: May 28-29, 2026

- Day 1:
 - 10:30 - 12:30 am
 - 30 minute break
 - 1:00 - 3:00 pm
- Day 2:
 - 10:30 - 12:30 am
 - 30 minute break
 - 1:00 - 3:00 pm

Workshop Resources

- GoogleDrive
 - See Statistical Horizons Course Link
- SLACK:
 - <https://r-for-sas-users-sh.slack.com>
- WEBSITE:
 - <https://tinyurl.com/r4sasusers>
- H.E.L.P. Dataset and “SAS and R” Book
 - <https://nhorton.people.amherst.edu/sasr2/datasets.php>
 - <https://nhorton.people.amherst.edu/sasr2/>

Compare R and SAS

R

- Bare Bones (89MB)
- Takes up little memory
- Powerful base packages included
- FREE/Open Source
- Add “packages” as you go – only “load” what you need at that time

SAS

- Have to buy base \$\$ & Large (20GB+)
- Almost always have to buy add-ons, gets expensive \$\$
- Must know what you want ahead of time
- Sometimes end up with more than you need

My Current SAS License Includes:

```

1 sascode_d1_01.sas
* =====;
* get details on SAS components and
  products installed;
* =====;

proc setinit; run;

Warning Period: 15 days (ending 29AUG2026).
System birthday: 08JUL2025.
Operating System:  Wx64_WKS.
Product expiration dates:
---Base SAS Software
  14AUG2026 (CPU A)
---SAS/STAT
  14AUG2026 (CPU A)
---SAS/GRAPH
  14AUG2026 (CPU A)
---SAS/ETS
  14AUG2026 (CPU A)
---SAS/FSP
  14AUG2026 (CPU A)
---SAS/OR
  14AUG2026 (CPU A)
---SAS/AF
  14AUG2026 (CPU A)
---SAS/IML
  14AUG2026 (CPU A)
---SAS/QC
  14AUG2026 (CPU A)
---SAS/SHARE
  14AUG2026 (CPU A)
---SAS/ASSIST
  14AUG2026 (CPU A)
---SAS/CONNECT
  14AUG2026 (CPU A)
---SAS/EIS
  14AUG2026 (CPU A)
---SAS/GIS
  14AUG2026 (CPU A)
---SAS/SHARE*NET
  14AUG2026 (CPU A)
---SAS Enterprise Miner
  14AUG2026 (CPU A)
---SAS Bridge for ESRI
  14AUG2026 (CPU A)
---OR OPT
  14AUG2026 (CPU A)
---OR PRS
  14AUG2026 (CPU A)
---OR IVS
  14AUG2026 (CPU A)
---OR LSO
  14AUG2026 (CPU A)
---SAS/ACCESS Interface to DB2
  14AUG2026 (CPU A)
---SAS/ACCESS Interface to Oracle
  14AUG2026 (CPU A)
---SAS/ACCESS Interface to SAP ASE
  14AUG2026 (CPU A)
---SAS/ACCESS Interface to PC Files
  14AUG2026 (CPU A)
---SAS/ACCESS Interface to ODBC
  14AUG2026 (CPU A)
---SAS/ACCESS Interface to OLE DB
  14AUG2026 (CPU A)
---SAS/ACCESS Interface to R/3
  14AUG2026 (CPU A)
---SAS/ACCESS Interface to Teradata
  14AUG2026 (CPU A)
---SAS/ACCESS Interface to Microsoft SQL Server
  14AUG2026 (CPU A)
---SAS/ACCESS Interface to MySQL
  14AUG2026 (CPU A)
---SAS Enterprise Miner for Desktop
  14AUG2026 (CPU A)
---SAS UML Studio
  14AUG2026 (CPU A)
---SAS Workspace Server for Local Access
  14AUG2026 (CPU A)
---SAS/ACCESS Interface to Netezza
  14AUG2026 (CPU A)
---SAS/ACCESS Interface to Greenplum
  14AUG2026 (CPU A)
---SAS/ACCESS Interface to SAP IQ
  14AUG2026 (CPU A)
---SAS/ACCESS to Hadoop
  14AUG2026 (CPU A)
---SAS/ACCESS to Vertica
  14AUG2026 (CPU A)
---SAS/ACCESS to Postgres
  14AUG2026 (CPU A)
---SAS/ACCESS to Impala
  14AUG2026 (CPU A)
---SAS/ACCESS to Salesforce
  14AUG2026 (CPU A)
---SAS/ACCESS to Amazon Redshift
  14AUG2026 (CPU A)
---High Performance Suite
  14AUG2026 (CPU A)
---SAS Add-in for Microsoft Excel
  14AUG2026 (CPU A)
---SAS/ACCESS to SAP HANA
  14AUG2026 (CPU A)
---PRODNUM0896
  14AUG2026 (CPU A)
---SAS/ACCESS Interface to the PI System
  14AUG2026 (CPU A)
---SAS/ACCESS to JDBC
  14AUG2026 (CPU A)
---prod 1312
  14AUG2026 (CPU A)

```

SAS Software Components

```
2 proc product_status; run;
```

For Base SAS Software ...
Custom version information: 9.4_M6
Image version information: 9.04.01M6P110718

* For SAS/STAT ...
Custom version information: 15.1

* For SAS/GRAPH ...
Custom version information: 9.4_M6

* For SAS/ETS ...
Custom version information: 15.1

* For SAS/FSP ...
Custom version information: 9.4_M6

* For SAS/OR ...
Custom version information: 15.1

* For SAS/AF ...
Custom version information: 9.4_M6

* For SAS/IML ...
Custom version information: 15.1

* For SAS/QC ...
Custom version information: 15.1

* For SAS/SHARE ...
Custom version information: 9.4_M5

* For SAS/ASSIST ...
Custom version information: 9.4
Image version information: 9.04.01M0P061913

* For SAS/CONNECT ...
Custom version information: 9.4_M6

* For SAS/ACCESS to JDBC ...
Custom version information: 1.0

* For SAS/EIS ...
Custom version information: 9.4_M6

* For SAS/GIS ...
Custom version information: 9.4_M6

* For SAS/ACCESS Interface to Netezza ...
Custom version information: 9.43

* For SAS/ACCESS Interface to Greenplum ...
Custom version information: 9.43

* For SAS/ACCESS Interface to SAP IQ ...
Custom version information: 9.4_M5

* For SAS/ACCESS to Hadoop ...
Custom version information: 9.46

* For SAS Enterprise Miner ...
Custom version information: 15.1

* For SAS/ACCESS to Vertica ...
Custom version information: 9.4_M6

* For SAS/ACCESS to Postgres ...
Custom version information: 9.4_M6

=====;

product_status; long word s. It whether use;

=====

For SAS/ACCESS to Impala ...
Custom version information: 9.45

For SAS/ACCESS to Salesforce ...
Custom version information: 9.4

Image version information: 9.04.01M6P041019

For SAS/ACCESS to Amazon Redshift ...
Custom version information: 9.43

For SAS Integration Technologies ...
Custom version information: 9.4_M6

For SAS Text Miner ...
Custom version information: 15.1

For High Performance Suite ...
Custom version information: 2.2_M7

For SAS/ACCESS Interface to DB2 ...
Custom version information: 9.4_M2

For SAS/ACCESS Interface to Oracle ...
Custom version information: 9.43

For SAS/ACCESS Interface to the PI System ...
Custom version information: 9.44

For SAS/ACCESS Interface to SAP ASE ...
Custom version information: 9.4_M5

For SAS/ACCESS Interface to PC Files ...
Custom version information: 9.4_M6

For SAS/ACCESS Interface to ODBC ...
Custom version information: 9.4_M6

For SAS/ACCESS Interface to OLE DB ...
Custom version information: 9.4_M6

For SAS/ACCESS Interface to Teradata ...
Custom version information: 9.46

For SAS/ACCESS Interface to MySQL ...
Custom version information: 9.4_M6

NOTE: PROCEDURE PRODUCT_STATUS used (Total process time):
real time 0.12 seconds
cpu time 0.11 seconds

My Current R - Base Installation

```
Console | Terminal x | Background Jobs x
R 4.6.0 · ~/
> sessionInfo()
R version 4.6.0 (2026-04-24 ucrt)
Platform: x86_64-w64-mingw32/x64
Running under: Windows 11 x64 (build 22000)

Matrix products: default
LAPACK version 3.12.1

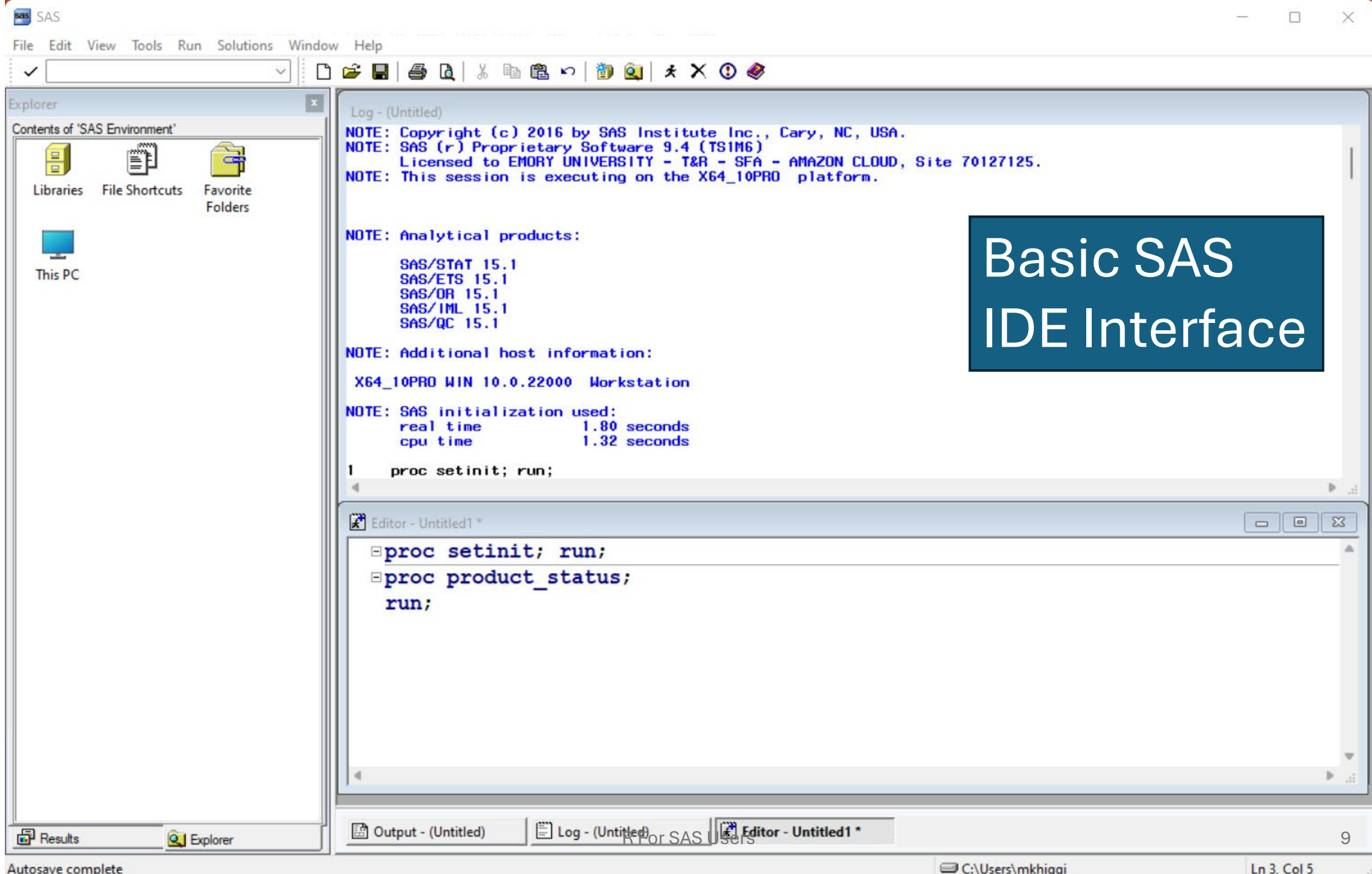
locale:
[1] LC_COLLATE=English_United States.utf8
[2] LC_CTYPE=English_United States.utf8
[3] LC_MONETARY=English_United States.utf8
[4] LC_NUMERIC=C
[5] LC_TIME=English_United States.utf8

time zone: America/New_York
tzcode source: internal

attached base packages:
[1] stats      graphics  grDevices  utils      datasets  methods
[7] base
R For SAS Users
```

Only 7 Base Packages

Interfaces – SAS Base; SAS Studio



SAS

File Edit View Tools Run Solutions Window Help

Results

Results

SGPlot: Horsepower vs. Weight

Log - (Untitled)

```
3 proc sgplot data=sashelp.cars;  
4 title "Horsepower vs. Weight";  
5 scatter x=weight y=horsepower / group=origin;  
6 reg x=weight y=horsepower;  
7 run;
```

NOTE: Writing HTML Body file: sashtml.htm
NOTE: PROCEDURE SGPLOT used (Total process time):
real time 1.35 seconds
cpu time 0.45 seconds
NOTE: There were 428 observations read from the data set SASHELP.

Output LOG

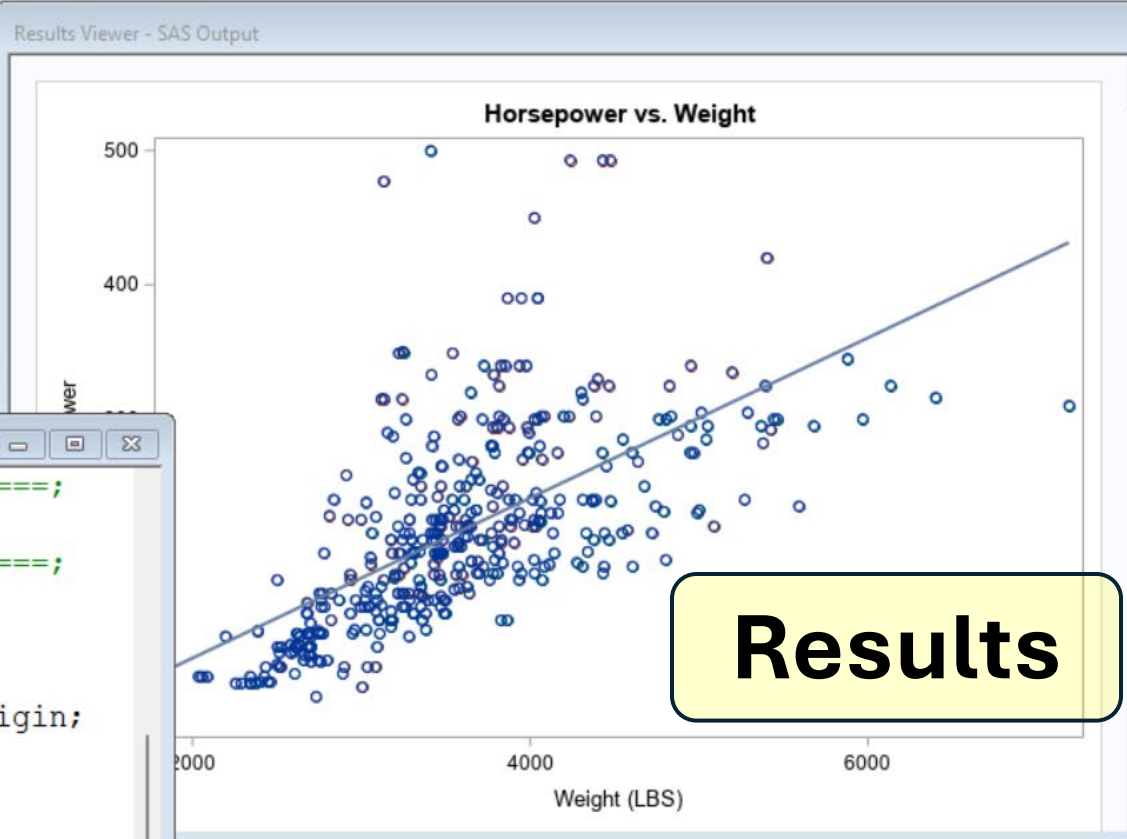
sascode_d1_01.sas *

```
* =====;  
* example plot using builtin cars dataset;  
* =====;  
  
proc sgplot data=sashelp.cars;  
title "Horsepower vs. Weight";  
scatter x=weight y=horsepower / group=origin;  
reg x=weight y=horsepower;  
run;
```

SAS Program Script

Results Viewer - SAS Output

Horsepower vs. Weight



Results

Output - (Untitled) Log - (Untitled) sascode_d1_01.sas * Results Viewer - SAS Ou...

NOTE: 5 Lines Submitted.

C:\Users\rmkhiggi Ln 28, Col 1

SAS Studio Interface IDE/GUI

localhost:63567/main?locale=en_US&zone=G

SAS® Studio

SAS Programmer Sign Out

Files and Folders

- about_files
- archived
- cleandemo
- docs
- images
- installR
- rstudiocustom
- rstudioide
- sascodes
 - sascode_d1_01.sas**
- slides
- _extensions
- _site
 - additionalResources.qmd
 - bioconductor_phyloseq.png
 - bioconductor_pkg.png
 - cc_by_nc_nd.png
 - compare_rggobi_tourr.png
 - compare_tablepkg.png

Tasks and Utilities

Snippets

Libraries

File Shortcuts

sascode_d1_01.sas

CODE LOG RESULTS

```
1 * =====;
2 * get details on SAS components and
3 * products installed;
4 * =====;
5
6 proc setinit; run;
7
8 * =====;
9 * returns a list of the SAS Foundation products
10 * that are installed on your system, along with
11 * the version numbers of those products. It
12 * provides a quick method to determine whether
13 * a SAS product is available for your use;
14 * =====;
15
16 proc product_status; run;
17
18 * =====;
19 * example plot using builtin dataset;
20 * =====;
21
22 proc sgplot data=sashelp.cars;
23     title "Horsepower vs. Weight";
24     scatter x=weight y=horsepower / group=origin;
25     reg x=weight y=horsepower;
26 run;
27
28
```

Line #

C:/MyGithub/StatisticalHorizons_R4SASUsers/sascodes/sascode_d1_01.sas

Line 28, Column 1 UTF-8

Messages: 1 User: Higgins, Melinda K

SAS Program Script

SAS Studio Interface IDE/GUI

localhost:63567/main?locale=en_US&zone=G

SAS® Studio

SAS Programmer Sign Out

Files and Folders

- about_files
- archived
- cleandemo
- docs
- images
- installR
- rstudiocustom
- rstudioide
- sascodes
 - sascode_d1_01.sas
- slides
- _extensions
- _site
 - additionalResources.qmd
 - bioconductor_phyloseq.png
 - bioconductor_pkg.png
 - cc_by_nc_nd.png
 - compare_rggobi_tourr.png
 - compare_tablepkg.png

Tasks and Utilities

Snippets

Libraries

File Shortcuts

sascode_d1_01.sas

CODE LOG RESULTS

Errors, Warnings, Notes

- Errors
- Warnings
- Notes (2)

Output LOG

```
1      OPTIONS NONOTES NOSTIMER NOSOURCE NOSYNTAXCHECK;
72
73      proc sgplot data=sashelp.cars;
74          title "Horsepower vs. Weight";
75          scatter x=weight y=horsepower / group=origin;
76          reg x=weight y=horsepower;
77      run;

NOTE: PROCEDURE SGPLOT used (Total process time):
      real time           2.98 seconds
      cpu time            1.12 seconds

NOTE: There were 428 observations read from the data set
      SASHELP.CARS.

78
79      OPTIONS NONOTES NOSTIMER NOSOURCE NOSYNTAXCHECK;
91
```

Messages: 1 User: Higgins, Melinda K

SAS Studio Interface IDE/GUI

← ↻ ⓘ localhost:63567/main?locale=en_US&zone=G A ☆ ☆ ... Chat

SAS® Studio SAS Programmer ? Sign Out

Files and Folders

- about_files
- archived
- cleandemo
- docs
- images
- installR
- rstudiocustom
- rstudioide
- sascodes
 - sascode_d1_01.sas**
- slides
- _extensions
- _site
 - additionalResources.qmd
 - bioconductor_phyloseq.png
 - bioconductor_pkg.png
 - cc_by_nc_nd.png
 - compare_rggobi_tourr.png
 - compare_tablepkg.png

Tasks and Utilities

Snippets

Libraries

File Shortcuts

sascode_d1_01.sas

CODE LOG RESULTS

Table of Contents

Results

Horsepower vs. Weight

Horsepower

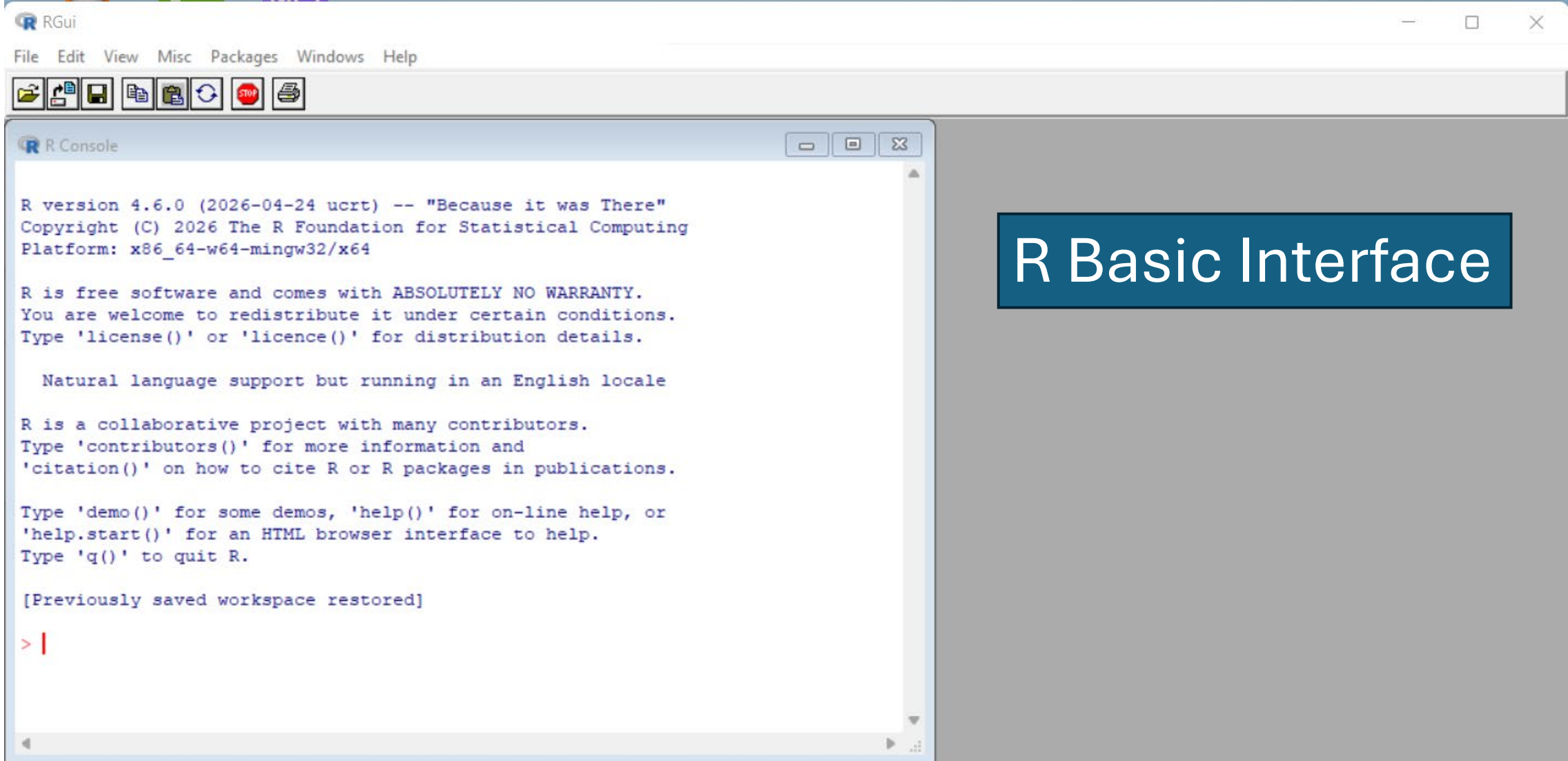
Weight (LBS)

Origin: Asia Europe USA

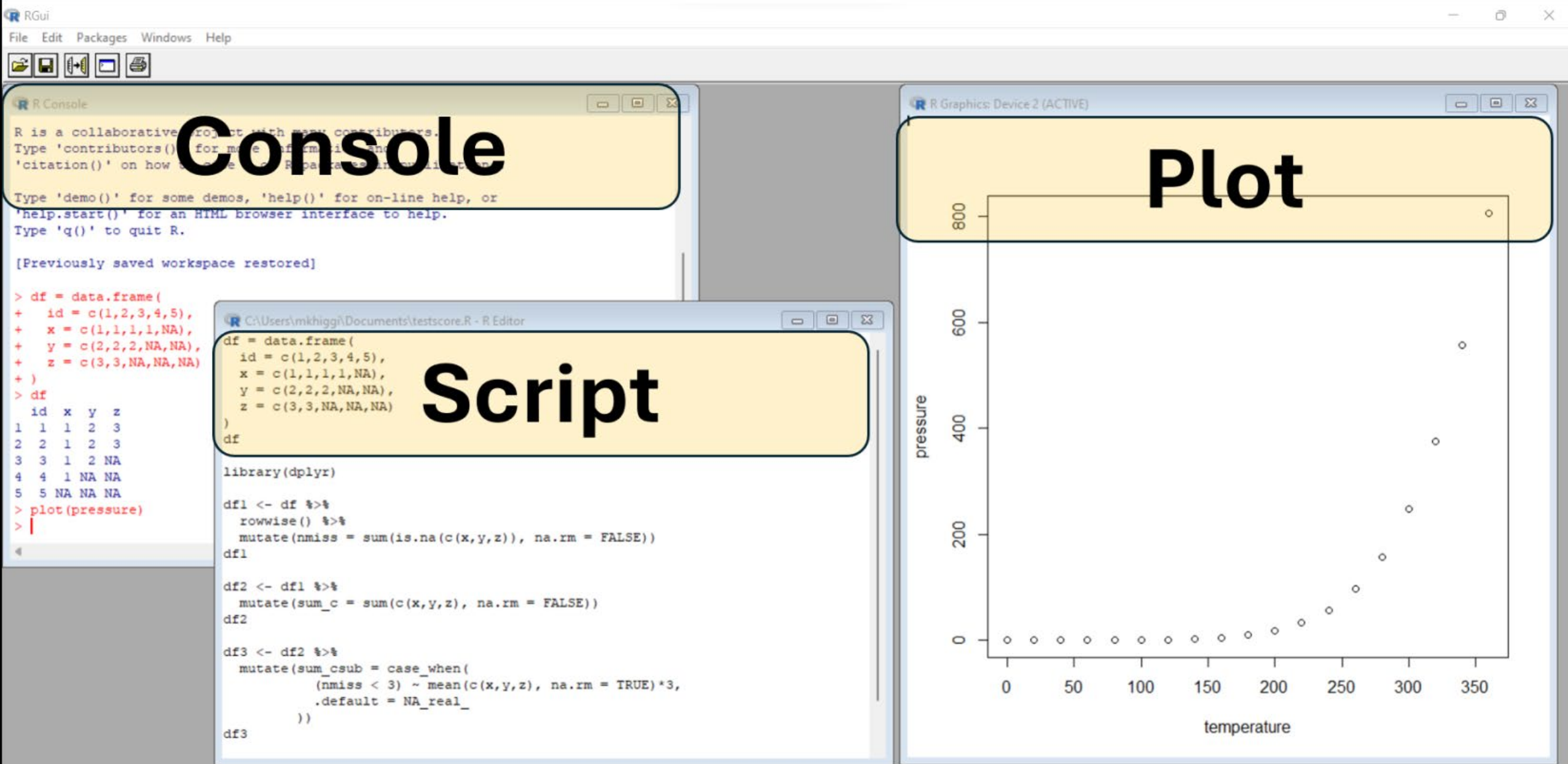
Messages: 1 User: Higgins, Melinda K

Interfaces – R Basic; Rstudio IDE; Positron

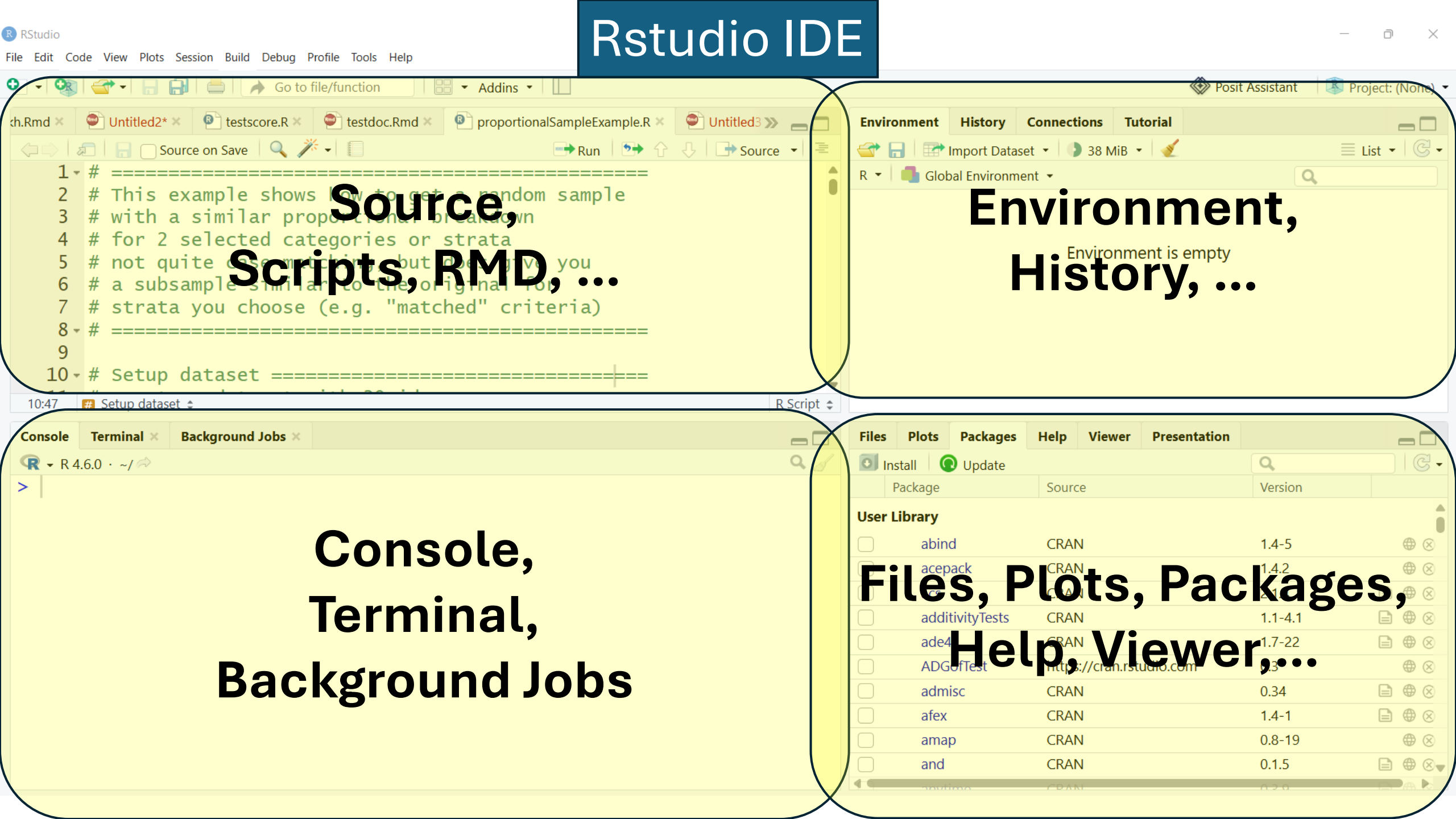
- R (from CRAN) – FREE – comes with a basic programming interface
 - <https://cran.r-project.org/>
- Rstudio is an IDE = Integrated Development Environment
 - <https://posit.co/download/rstudio-desktop>
- Positron is the next evolution of programming interfaces
 - <https://positron.posit.co/>
 - Learn more, <https://positron.posit.co/migrate-rstudio-compare.html>



R Basic Interface



R Basic Interface – similar to SAS Basic Interface



Rstudio IDE

**Source,
Scripts, RMD, ...**

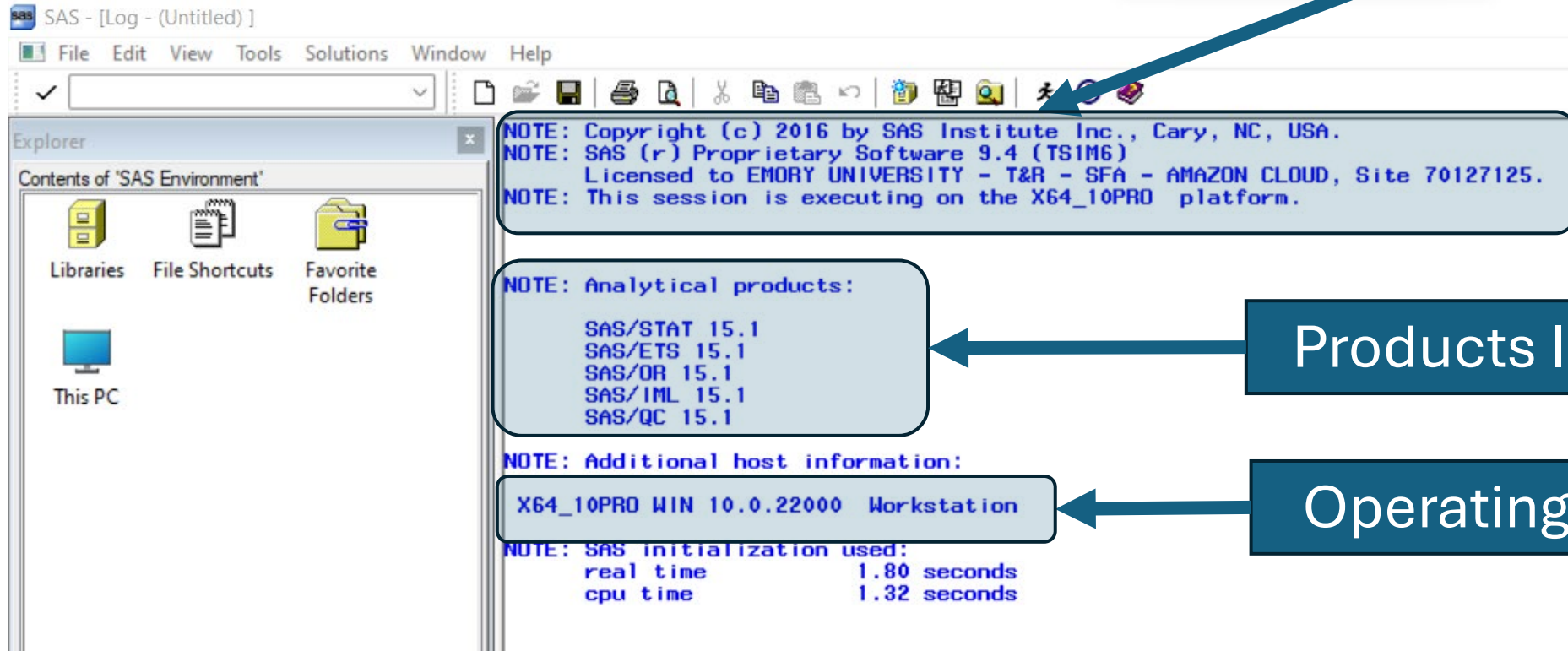
**Environment,
History, ...**

**Console,
Terminal,
Background Jobs**

**Files, Plots, Packages,
Help, Viewer, ...**

SAS Session

SAS Version and
Local System Details



The screenshot shows the SAS software interface. The title bar reads "SAS - [Log - (Untitled)]". The menu bar includes "File", "Edit", "View", "Tools", "Solutions", "Window", and "Help". The toolbar contains various icons for file operations and execution. On the left, the "Explorer" pane shows the "Contents of 'SAS Environment'" with icons for "Libraries", "File Shortcuts", "Favorite Folders", and "This PC". The main log window displays the following text:

```
NOTE: Copyright (c) 2016 by SAS Institute Inc., Cary, NC, USA.  
NOTE: SAS (r) Proprietary Software 9.4 (TS1M6)  
       Licensed to EMORY UNIVERSITY - T&R - SFA - AMAZON CLOUD, Site 70127125.  
NOTE: This session is executing on the X64_10PRO platform.  
  
NOTE: Analytical products:  
      SAS/STAT 15.1  
      SAS/ETS 15.1  
      SAS/OR 15.1  
      SAS/IML 15.1  
      SAS/QC 15.1  
  
NOTE: Additional host information:  
      X64_10PRO WIN 10.0.22000 Workstation  
  
NOTE: SAS initialization used:  
      real time      1.80 seconds  
      cpu time       1.32 seconds
```

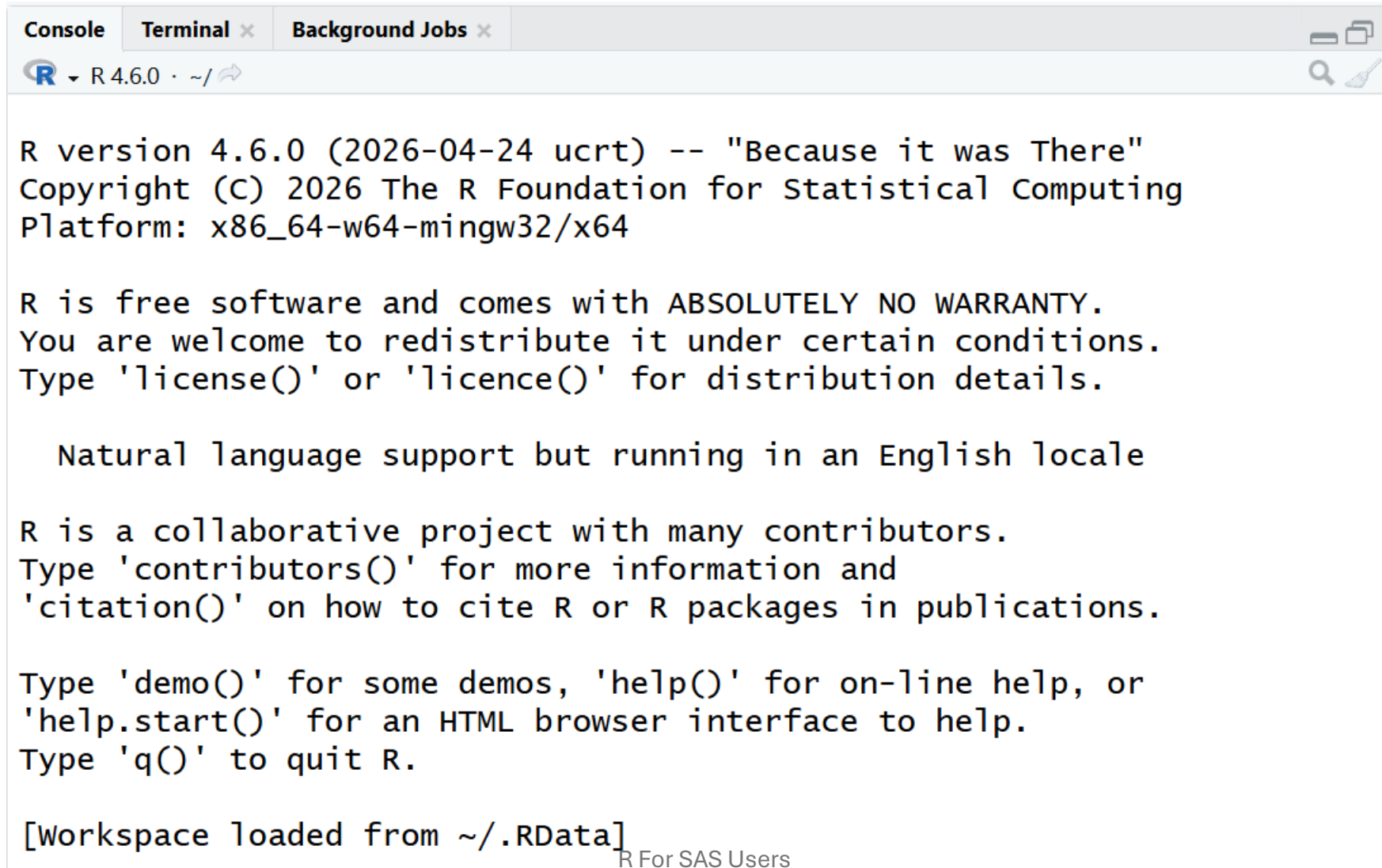
Four callout boxes with arrows point to specific parts of the log:

- SAS Version and Local System Details** points to the first three lines of the log.
- Products Included** points to the "Analytical products" section.
- Operating System** points to the "Additional host information" section.
- A fourth callout box (not explicitly labeled) points to the "SAS initialization used" section.

Products Included

Operating System

R Session – Details on Start

A screenshot of an R console window. The window has a title bar with tabs for 'Console', 'Terminal', and 'Background Jobs'. The 'Console' tab is active. The console shows the R version 4.6.0 (2026-04-24 ucrt) and the platform x86_64-w64-mingw32/x64. It displays the R startup message, including the copyright notice and the license information. The console also shows the workspace loaded from ~/.RData.

```
R version 4.6.0 (2026-04-24 ucrt) -- "Because it was There"
Copyright (C) 2026 The R Foundation for Statistical Computing
Platform: x86_64-w64-mingw32/x64

R is free software and comes with ABSOLUTELY NO WARRANTY.
You are welcome to redistribute it under certain conditions.
Type 'license()' or 'licence()' for distribution details.

Natural language support but running in an English locale

R is a collaborative project with many contributors.
Type 'contributors()' for more information and
'citation()' on how to cite R or R packages in publications.

Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.

[Workspace loaded from ~/.RData]
```

R Session – sessionInfo()

R Version

```
Console | Terminal x | Background Jobs x
R 4.6.0 · ~/
> sessionInfo()
R version 4.6.0 (2026-04-24 ucrt)
Platform: x86_64-w64-mingw32/x64
Running under: Windows 11 x64 (build 22000)

Matrix products: default
LAPACK version 3.12.1

locale:
 [1] LC_COLLATE=English_United States.utf8
 [2] LC_CTYPE=English_United States.utf8
 [3] LC_MONETARY=English_United States.utf8
 [4] LC_NUMERIC=C
 [5] LC_TIME=English_United States.utf8

time zone: America/New_York
tzcode source: internal

attached base packages:
[1] stats      graphics  grDevices  utils      datasets  methods
[7] base
```

Operating System

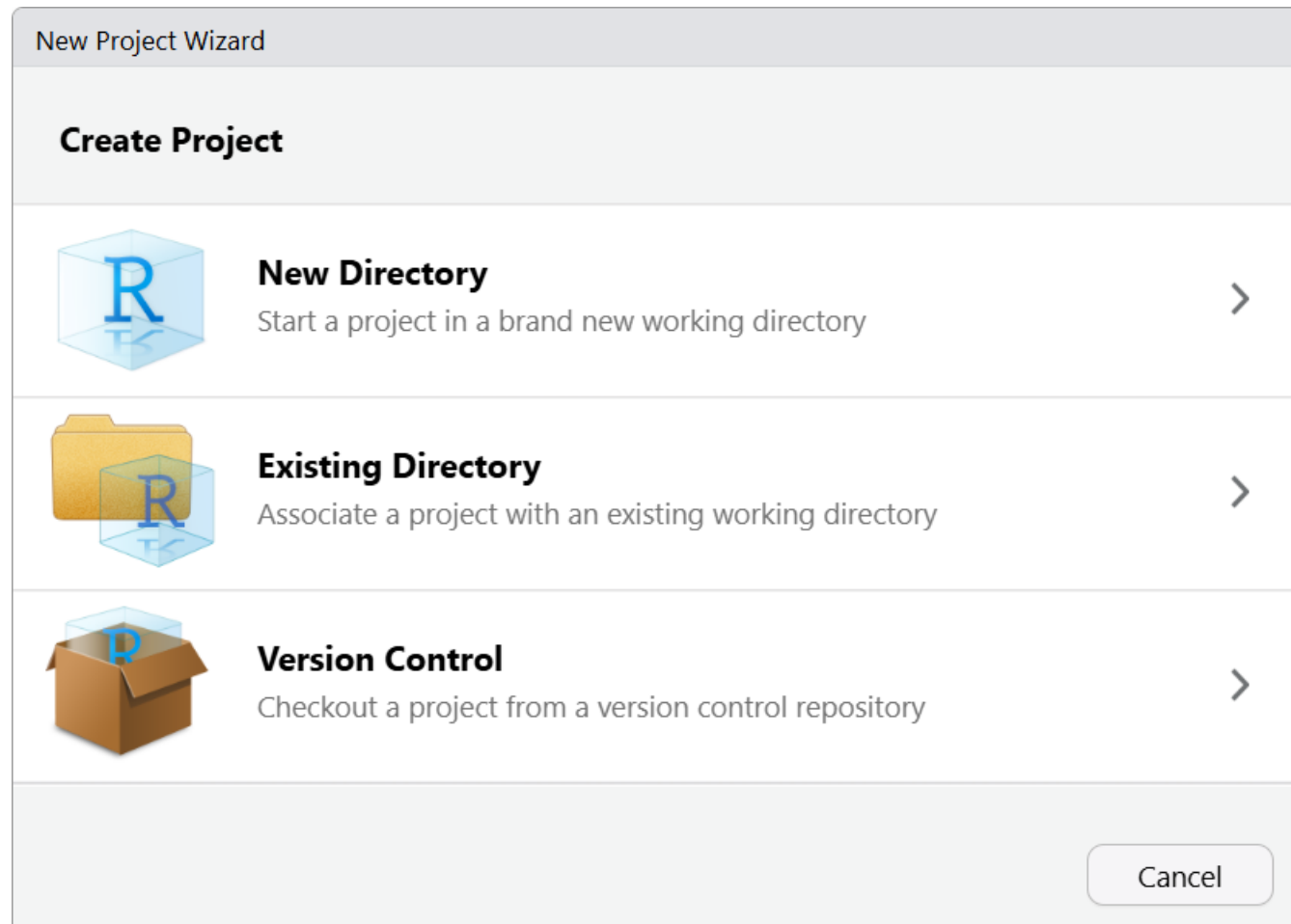
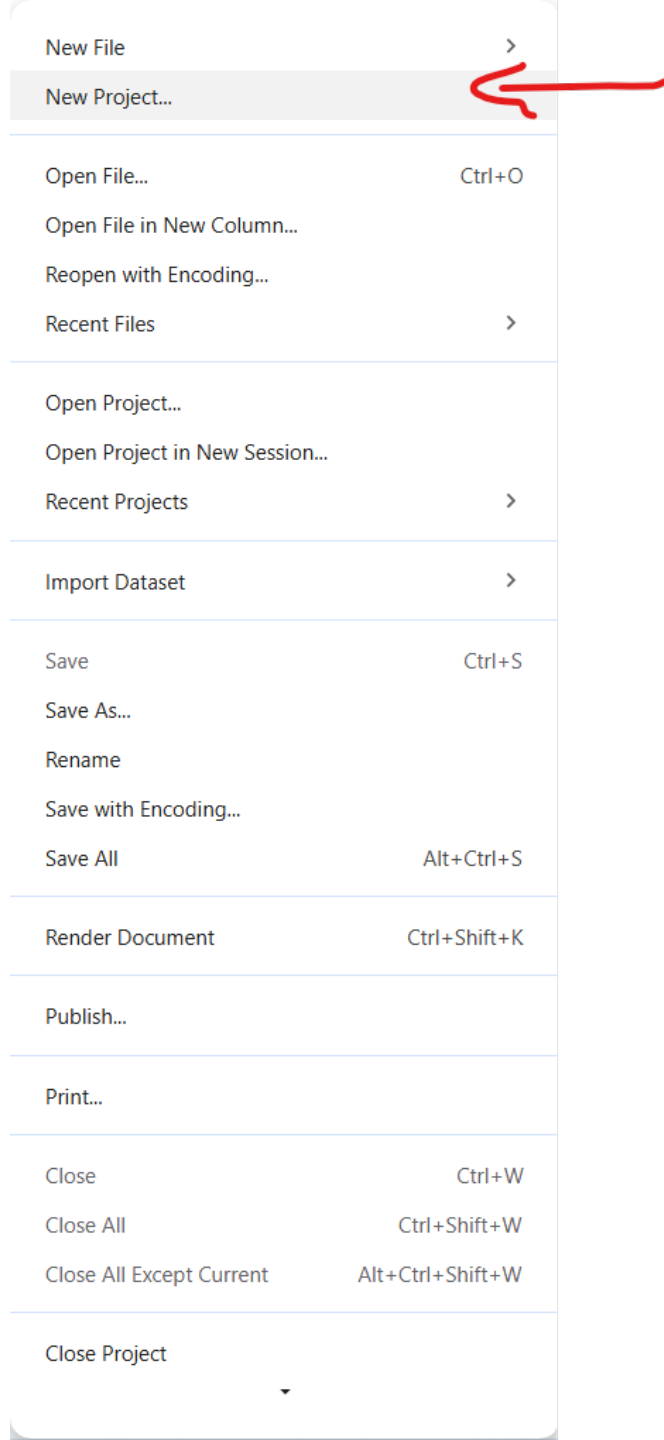
Local Details

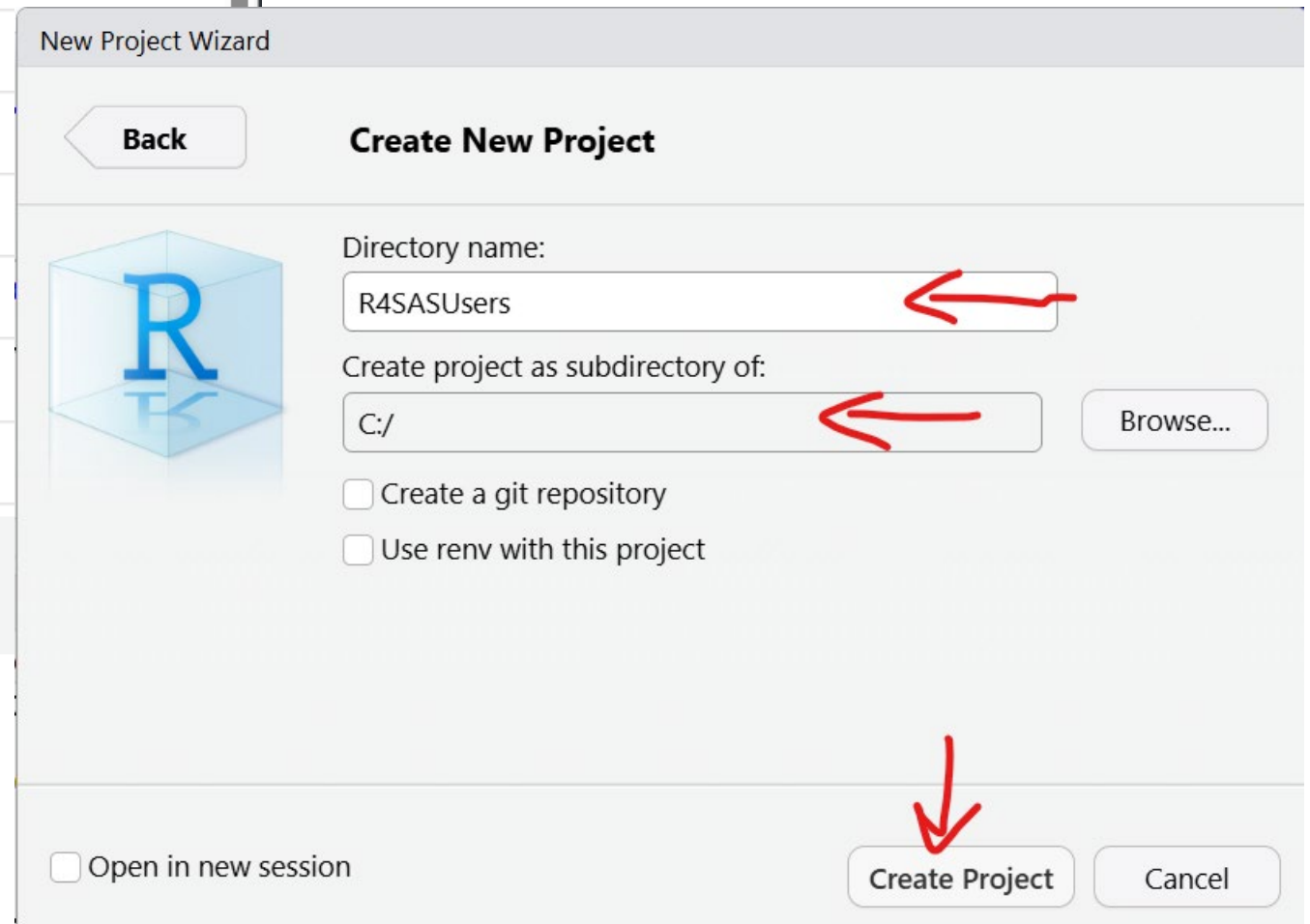
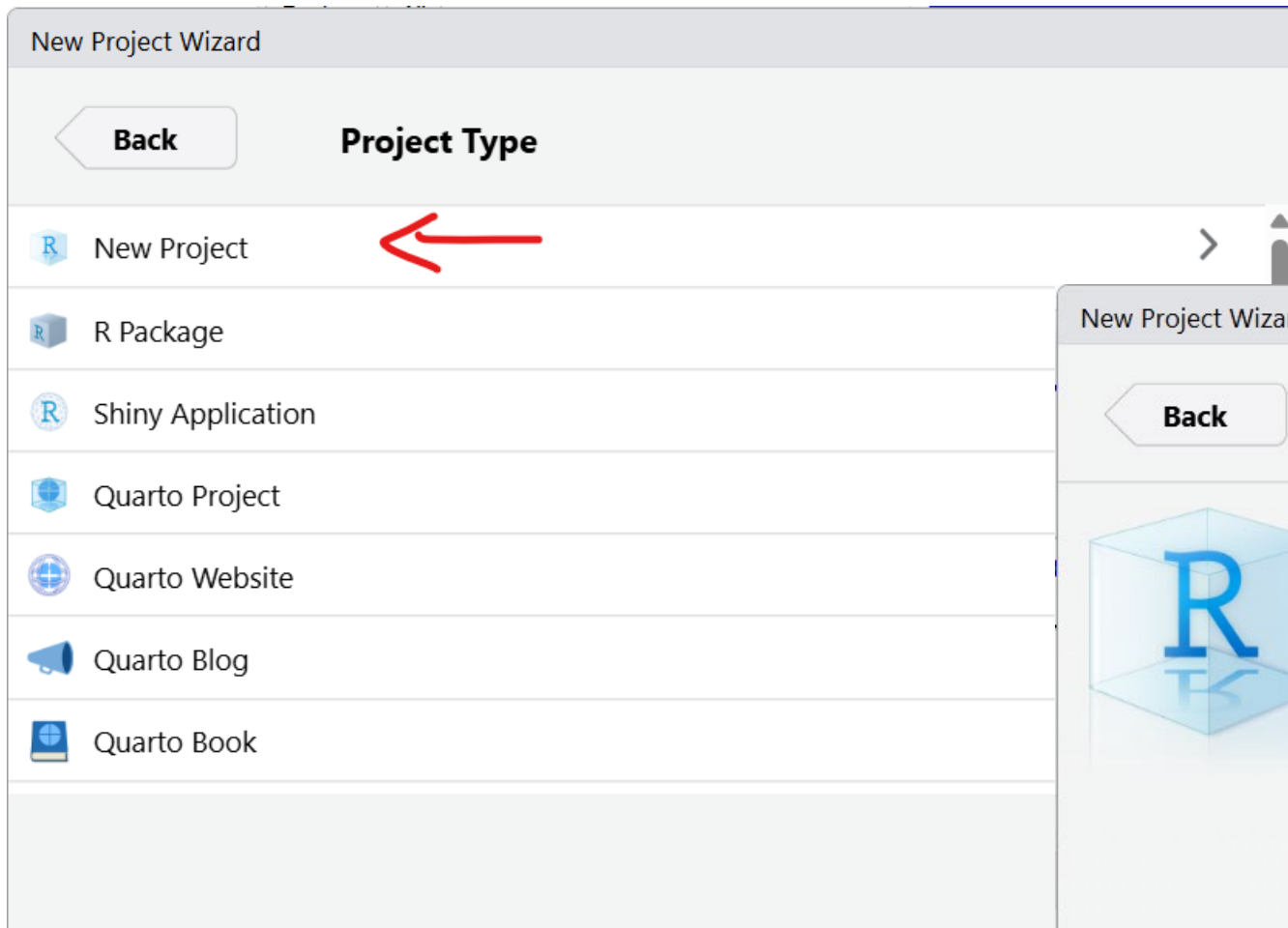
7 Base Packages

Open RStudio

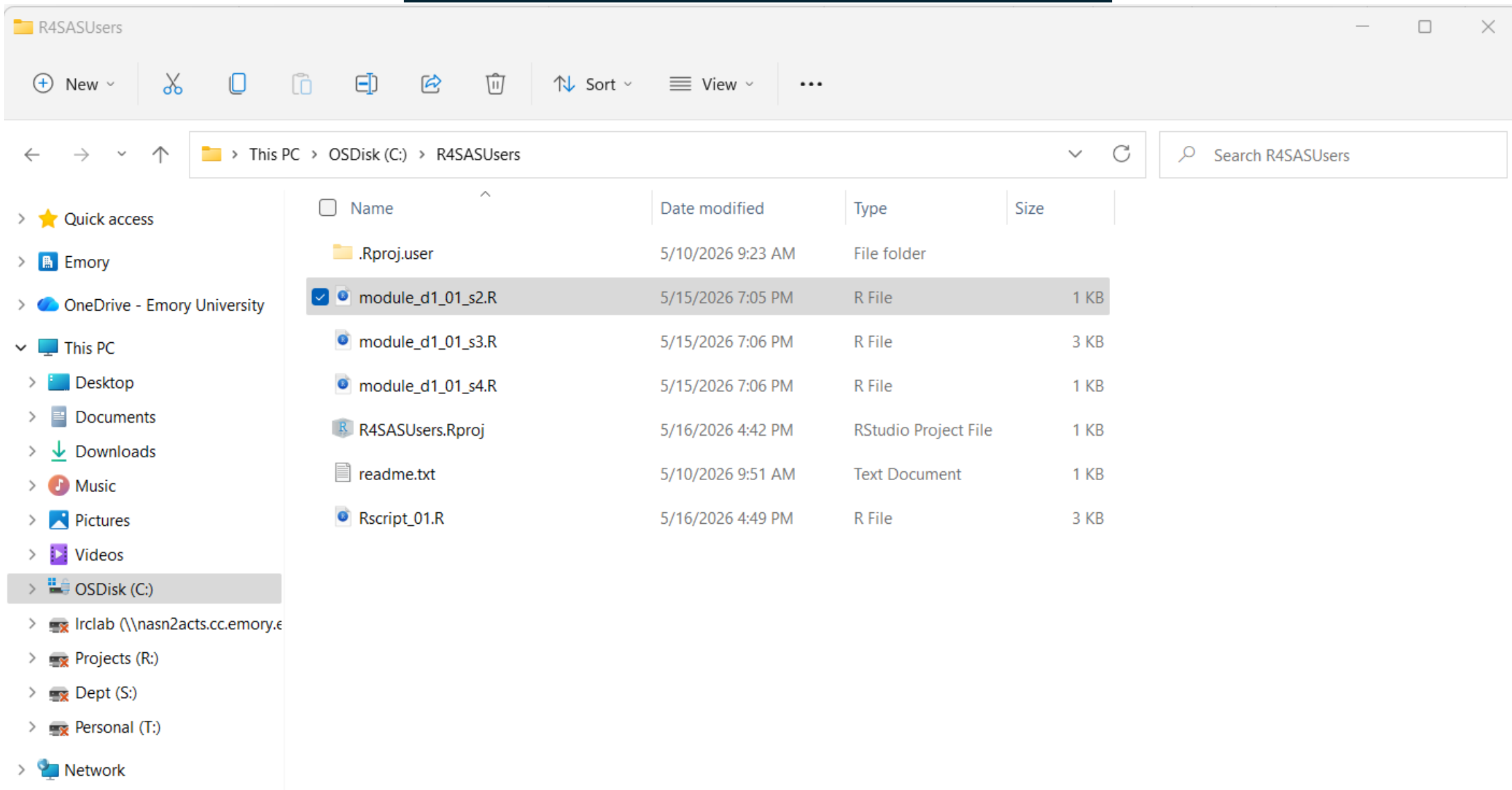
Exercise 1:

- Look at session details
- Create new project “R4SASUsers”: File/New Project
- Put all files for R For SAS Users into this folder





Put Files Into This Folder



The screenshot shows a Windows File Explorer window titled "R4SASUsers". The address bar indicates the path: "This PC > OSDisk (C:) > R4SASUsers". The left sidebar shows the "This PC" section expanded, with "OSDisk (C:)" selected. The main pane displays a list of files and folders:

Name	Date modified	Type	Size
.Rproj.user	5/10/2026 9:23 AM	File folder	
☒ module_d1_01_s2.R	5/15/2026 7:05 PM	R File	1 KB
module_d1_01_s3.R	5/15/2026 7:06 PM	R File	3 KB
module_d1_01_s4.R	5/15/2026 7:06 PM	R File	1 KB
R4SASUsers.Rproj	5/16/2026 4:42 PM	RStudio Project File	1 KB
readme.txt	5/10/2026 9:51 AM	Text Document	1 KB
Rscript_01.R	5/16/2026 4:49 PM	R File	3 KB

R4SASUsers - RStudio

File Edit Code View Plots Session Build Debug Profile Tools Help

Go to file/function

Posit Assistant R4SASUsers

Source

Environment History Connections Tutorial

Files Plots Packages Help Viewer Presentation

Folder File Delete Rename

C: > R4SASUsers

	Name	Size	Modified
	..		
	R4SASUsers.Rproj	217 B	May 16, 2026, 4:42 PM
	readme.txt	229 B	May 10, 2026, 9:51 AM
	module_d1_01_s2.R	921 B	May 15, 2026, 7:05 PM
	module_d1_01_s3.R	2.1 KB	May 15, 2026, 7:06 PM
	module_d1_01_s4.R	610 B	May 15, 2026, 7:06 PM
	Rscript_01.R	2.1 KB	May 16, 2026, 4:49 PM

R version 4.6.0 (2026-04-24 ucrt) -- "Because it was There"
Copyright (C) 2026 The R Foundation for Statistical Computing
Platform: x86_64-w64-mingw32/x64

R is free software and comes with ABSOLUTELY NO WARRANTY.
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Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.

[Workspace loaded from C:/R4SASUsers/.RData]

```
> getwd()
[1] "C:/R4SASUsers"
> |
```

Project Folder

R For SAS Users

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